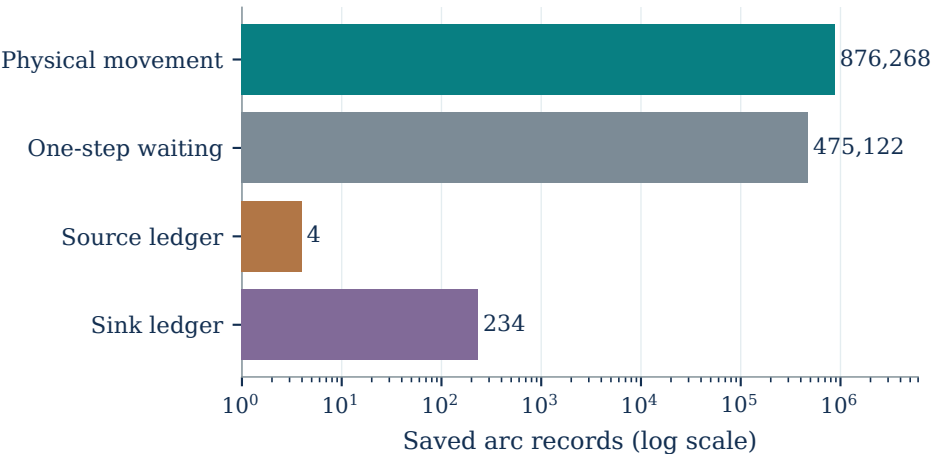
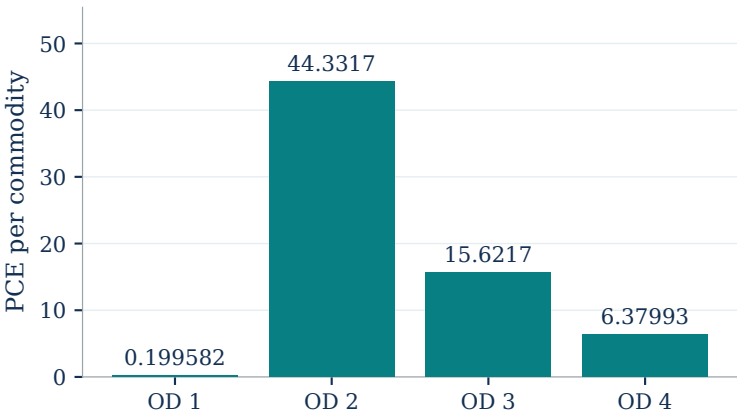


T4 construction: complete arc ledger and four-OD pulse

a All dynamic arcs



b Frozen five-minute demand pulse



1,351,628 arcs | 4 commodities | total q = 66.53295455 PCE

30 s per step; departure at step 5 (08:02:30); H = 169 steps (84.5 min after 08:00).

q = hourly Stage-3 demand / 12. This is a synthetic engineering pulse, not observed departures.

- OD 1 osm:9913425856 → osm:261207373
- OD 2 osm:261881717 → osm:9913425856
- OD 3 osm:5844620237 → osm:261881717
- OD 4 osm:9708251508 → osm:9913425856

Physical duration = ceil(free-flow minutes / 0.5), at least one step; cost = original free-flow minutes.
Road capacity = hourly PCE / 120. Waiting advances one step and costs 0.5 minute.
New turn sidecar excluded: this graph has no independent zero-time turn arcs.
Source/sink connectors are zero-cost bookkeeping; a jump to H is not physical waiting.